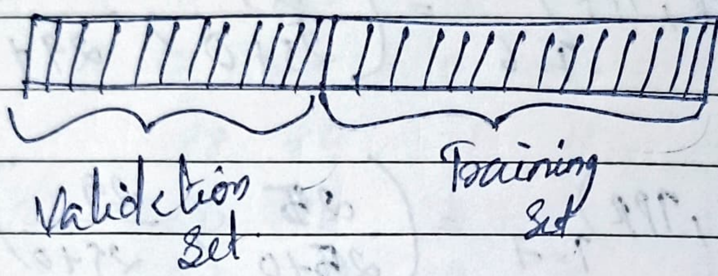
Cross Validation

To test the performance of a classifier, we need to have a number of training / validation set pairs.



Cross Validation methods are used for generating multiple training validation sets from given dataset.

✓ It is a technique to evaluate the predictive models by partitioning the original sample into a training set to train the model and a test set to evaluate it.

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Methods used

not applicable for small dataset

① Hold - out
② K - fold cross validation
③ Leave - one - out cross validation (LOOCV)

④ Bootstrapping.

Bootstrapping

In small dataset, generate multiple samples from it.

eg:- Let dataset be. A, B, C, D, E

Training set | Validation set

1st iteration A CDCAB | E

4 samples with repetition | remaining sample.

called sampling with replacement

2nd iteration

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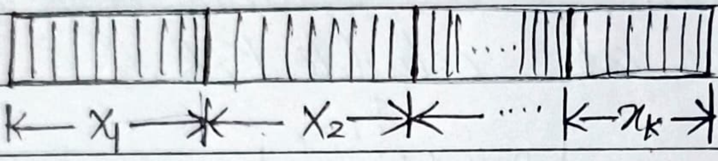
Bootstrapping is the process of computing performance measures using several randomly selected training and test datasets which are selected through a process of sampling with replacement.

Bootstrap procedure will create one or more new training datasets some of which are repeated.

The corresponding test datasets are then constructed from the set of examples that were not selected for the respective training datasets.

K-fold cross validation

The dataset X is divided into k-equal sized parts x_i, $i = 1, 2, \ldots, k$.



To generate each pair for testing and validation keep one of the k parts out as the validation set V_i and the remaining $(K-1)$ parts to form the training sets.

Doing this k times we get k pairs (V_i, T_i)

1st iteration $V_1 = X_1$ $\quad T_1 = X_2 \cup X_3 \cup \cdots \cup X_K$

2nd iteration $V_2 = X_2$ $\quad T_2 = X_1 \cup X_3 \cup X_4$

$\vdots$

K iteration $V_K = X_K$ $\quad T_K = X_1 \cup X_2 \cup \cdots \cup X_{K-1}$

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Bagging → Bootstrap + aggregating.

Boosting → Classify the samples.
↓
Incorrectly classify (Weightup)
↓
Continue until you get desired no. of models.

Bagging	Boosting
Builds the model parallely Given subset of original dataset	Builds the model sequentially Gives same dataset
Samples are unweighted	Samples are weighted.